

Alex Chiluisa

Ph.D. in Robotics **Engineering** | alexchiluisa@gmail.com | (+1) 508-410-0579 | Natick, Massachusetts

SUMMARY

I am an **organized** and self-motivated robotics engineer with a strong foundation in **mechanical** design and computational analysis, particularly focusing on their application in medicine. My expertise includes developing prototypes for electro-**mechanical** and robotic systems using SolidWorks, and employing 3D printing and CAM processes for rapid prototyping. I am skilled in MATLAB and Python for characterizing, simulating, and evaluating complex systems. I enjoy solving complex **engineering** challenges and work effectively in collaborative, multidisciplinary environments driving product development while ensuring adherence to **quality** standards and regulatory requirements.

EDUCATION

Worcester Polytechnic Institute, USA

May. 2024

Ph.D. Robotics **Engineering**

Worcester Polytechnic Institute, USA

Aug. 2018 - May. 2020

M.Sc. Robotics **Engineering**

Universidad de las Fuerzas Armadas - ESPE, Ecuador

Feb. 2008 - Sep. 2014

Bachelor of **Engineering**, Mechatronics.

TECHNICAL SKILLS

Hands-on experience in **mechanical** and robotics laboratories, involving the **mechanical** design and fabrication of prototypes, the execution of experiments, and the collecting and analysis of data.

Modeling and Analysis Tools

Solidworks, Finite Element Analysis, AutoCAD, **Mechanical** and Robotic Design, Statistical Analysis, 3D slicer.

Software Programming

Matlab, Simulink, Python, C++, PLC.

Hands-on Prototyping

3D Printing (FDM, SLA, and SLS), Medical Device Design, **CNC Milling Machining**, Laser Cutting, Robotic Hardware Development, Test Fixtures Design, PCB design.

RESEARCH EXPERIENCE

Worcester Polytechnic Institute, USA

Aug. 2020

Ph.D. - Research Assistant at Cognitive Medical Technology Laboratory

- Designed, modeled, and developed miniaturized flexible surgical instruments using Solidworks, enhancing tissue coverage by 100% in endoscopic laser surgeries.
- Established and implemented advanced micro-**CNC machining** methods, utilizing SolidWorks CAD/CAM for the internal production of precision components for medical devices following **GD&T** standard, improving production efficiency and reducing lead times.
- Designed and manufactured fixtures utilizing various 3D printing technologies, including FMD, SLA, and SLS, for experimental testing and validation of medical devices.
- Developed a comprehensive MATLAB framework to simulate and evaluate the operation of a flexible robotic system, optimizing design iterations and significantly accelerating the manufacturing process by reducing prototyping time and costs.
- Conducted kinematic characterization experiments to optimize the design, performance, and reliability of flexible instruments, utilizing custom Python scripts for data analysis.
- Designed, planned, and executed user studies to assess flexible surgical instruments' operation in real clinical scenarios.
- Mentored undergraduate students on research projects.
- Managed the collaboration with clinicians, engineers, and researchers to optimize and advance the development of surgical instruments.
- Coordinated daily operations as Lab Manager, supporting research projects by managing procurements and ensuring equipment maintenance.

Intern Research and Development in partnership with Boston Scientific

- Developed kinematic models for flexible endoscopes used in urology procedures to understand and optimize performance.
- Designed, implemented, and simulated PID controllers in Simulink to **control** a rigid manipulator and a ureteroscope concurrently, achieving precise motion coordination.
- Designed and assembled custom circuit boards to **control** DC motors, enabling precise motorization of a ureteroscope for improved maneuverability and functionality.
- Performed Finite Element Analysis (FEA) to optimize the design of a ureteroscope holder, ensuring precise positioning and stability for accurate and effective instrument operation.
- Tested and troubleshoot electronic devices using laboratory equipment such as oscilloscopes and digital equipment.

WORK EXPERIENCE**Worcester Polytechnic Institute, USA**

Aug. 2018 - May 2020

Teaching Assistant

- Provided academic support to students by reinforcing concepts and techniques taught in Introduction to Robotics and Robot Dynamics courses, including hands-on guidance during laboratory sessions.
- Evaluated assignments and maintained records of course materials, ensuring consistency and clarity in grading and lesson documentation.

Procesadora VYMSA, Ecuador

Dec. 2014 - Dec. 2015

Design Engineer

- Engineered and optimized components for white goods appliances using SolidWorks and AutoCAD, introducing new products into the **assembly** line.
- Developed and implemented manufacturing protocols to integrate new products into serial production.
- Established and managed Rapid Prototyping and **CNC machining** areas, with expertise in equipment and process validation (IQ, OQ, and PQ) to ensure regulatory compliance and operational efficiency, accelerating the early-stage development and proof-of-concept phases for multiple projects.
- Managed the calibration and maintenance of measuring instruments, including calipers, scales, and force gauges, ensuring accuracy and **quality control** across production processes.

PUBLICATIONS**Journal Papers**

1. DeVore, E., **Chiluisa, A.J.**, Minch, E.V., Mihaleva, R., Do, H., Tougas, R., Fichera, L. and Carroll, T.L. Benefits of Side-Firing Optical Fibers in Endoscopic Laser Treatment of the Larynx. *The Laryngoscope*, 2022.

Conference Papers

1. **Chiluisa, A.J.**, Pacheco, N.E., Do, H.S., Tougas, R.M., Minch, E.V., Mihaleva, R., Shen, Y., Liu, Y., Carroll, T.L. and Fichera, L. Light in the Larynx: a Miniaturized Robotic Optical Fiber for In-office Laser Surgery of the Vocal Folds. In 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 427-434. IEEE, 2022.
2. I.A. Chan, J.F. d'Almeida, **A.J. Chiluisa**, T.L. Carroll, Y. Liu, L. Fichera, On the Merits of Using Angled Fiber Tips in Office-based Laser Surgery of the Vocal Folds, in *SPIE Medical Imaging*, San Diego, CA, USA, 2021.
3. M. Zhu, Y. Shen, **A.J. Chiluisa**, J. Song, L. Fichera, Y. Liu, Optical Fiber Coupling System for Steerable Endoscopic Instruments *IEEE Engineering in Medicine And Biology Conference*, 2021.
4. **A.J. Chiluisa**, F.J. Van Rossum, J.B. Gafford, R.F. Labadie, R.J. Webster III, L. Fichera, Computational Optimization of Notch Spacing for a Transnasal Ear Endoscopy Continuum Robot, in *International Symposium on Medical Robotics*, Georgia, USA, 2020.

Thesis

1. **Chiluisa, Alex.**, New Miniaturized Wristed Instruments for In-office Endoscopic Laser Surgery. Ph.D. Dissertation. Worcester Polytechnic Institute, 2024.
2. Almache, D.R. and **Chiluisa, A.J.**, Design and construction of an automatic baler for compacting cardboard, paper and PET for up to 30 tons. Bachelor's degree Dissertation to become Mechatronic Engineer, Department of Energy Sciences and Mechanics, Universidad de las Fuerzas Armadas, Sangolqui, Ecuador, Oct 2014.

Alex Chiluisa

- Natick, MA, US

Contact Information

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- 5084100579

Skills

- **milling** - 5 years
- **medical imaging** - 1 years
- **cad** - 10 years
- **collaboration** - 10 years
- **engineering** - 10 years
- **matlab** - 10 years
- **solidworks** - 10 years
- **cam** - 8 years
- **3d printing** - 5 years
- **cnc** - 5 years
- **data analysis** - 5 years
- **equipment maintenance** - 5 years
- **laboratory management** - 5 years
- **leadership** - 5 years
- **machining** - 5 years
- **manufacturing** - 5 years
- **medical devices** - 5 years
- **operations** - 5 years
- **production** - 5 years
- **qa** - 5 years
- **research** - 5 years
- **robotics** - 5 years
- **lasers** - 4 years
- **laboratory equipment** - 5 years
- **pid** - 5 years
- **documentation** - 10 years
- **training** - 5 years
- **verification and validation** - 5 years

Work Preferences

- Desired Work Settings: Hybrid
- Likely to Switch: True
- Willing to Relocate: False
- Travel Preference: 25%
- Work Authorization:
 - US
- Work Documents:
 - Employment Auth. Document
- Security Clearance: False
- Third Party: False
- Employment Type:
 - Full-time

Profile Sources

- Dice:
<https://www.dice.com/employer/talent/profile/210dd7b4d13da1d852aab41e56955f38>